# ACE Troubleshooting Decision Tree

*ACE-DT-006*

AUTHOR Alex Morgan (fictional) REVIEW Instructor demonstration STATUS Evidence-led web access triage

## Purpose and assessment context

A structured instructor-authored model for CIT-205. It illustrates how technical work becomes verifiable evidence. All incidents, measurements, approvals and names below are fabricated for teaching.

### Start condition

Report: 'I cannot open the training website.' Capture URL, exact message, affected users, time, and device state.

### Decision A

Can other known websites open? YES: isolate the target site. NO: check link, IP configuration, gateway, and broader network.

### Decision B

Can the approved target IP be reached? YES and name fails: investigate DNS path. NO: investigate route, service, or filtering with approved diagnostics.

### Decision C

Does a different approved device reproduce the symptom? YES: escalate as possible shared service issue. NO: continue authorized local device investigation.

### Stop conditions

Unknown authorization, suspected security incident, change needed outside scope, or multiple impacted users: stop and escalate.

### Close condition

Reproduce initial failure, document hypothesis and test, perform approved fix, retest, confirm with user, and provide rollback notes.

### Evidence trail

Record observed signals and precise branch selected. Decision branches are hypotheses until supported by testing.

## Evidence and verification matrix

| DECISION | YES PATH | NO PATH | RECORD |
|---|---|---|---|
| **Other sites load?** | Target-specific | Local/network | Browser test |
| **Target IP works?** | DNS investigation | Routing/service | IP response |

| | | | |
|---|---|---|---|
| **Other device affected?** | Shared issue | Local issue | Comparison |
| **Fix authorized?** | Proceed & retest | Escalate | Approval |

# Reviewer handoff and quality assurance

Document: ACE-DT-006 | Revision: 1.2 | Classification: public instructional example

## Acceptance checklist

- ☐ Relevant user / system impact is defined without publishing confidential information.
- ☐ Expected outcomes are distinguished from the technician's actual observations.
- ☐ Every conclusion is supported by a specific test or appropriate evidence reference.
- ☐ Changes are authorized, reversible when appropriate, and documented.
- ☐ Closure includes verification, remaining risk, escalation, and reviewer disposition.
- ☐ Accessible language is used; screenshots and supporting files are safe for public release.

## Evidence register (model references)

ACE-DT-006 E-01 | Initial report or baseline. Synthetic exhibit reference, supplied to model documentation. Students replace with genuine, redacted records.

ACE-DT-006 E-02 | Troubleshooting observations. Synthetic exhibit reference, supplied to model documentation. Students replace with genuine, redacted records.

ACE-DT-006 E-03 | Change / review / authorization notes. Synthetic exhibit reference, supplied to model documentation. Students replace with genuine, redacted records.

ACE-DT-006 E-04 | Final validation and user confirmation. Synthetic exhibit reference, supplied to model documentation. Students replace with genuine, redacted records.

## Instructor review

What works: the record captures context, action, evidence and outcome. Before qualifying real student work, reviewers must inspect supporting artifacts and verify that any technical claim or metric is true.

Fictional review disposition: EXEMPLAR APPROVED FOR TEACHING ONLY.

## Reflection and next action

The value of this artifact is not the final checkbox; it is the trace from an observed symptom through a defensible decision to validated results. A real student should explain one revision made after feedback and link the relevant commit.